Case/Lab Identifier: Case Study 2 Reconstructing Chrome Web History
Trainee Name: Oluyede Tobi Steven
Registration Number: 202513549PF
Course: SBT-DF204 Computer Forensics Case Studies
Date: 12 October 2026
Instructor: Aminu Idris, AMCPN
Environment: Isolated Virtual Laboratory

1. EXECUTIVE CONCLUSION

This investigation examined a supplied Google Chrome History SQLite database to reconstruct relevant browser activity and establish a time-ordered sequence of recorded visits and download events.

The examination covered the database structure, URL records, visit timestamps, navigation transitions, previous-visit relationships, and available download artefacts. The findings were based on the records identified in the supplied database and documented in the accompanying SQL output, screenshots, timeline, and evidence log.

2. OBJECTIVES

The objectives of this investigation were to:

1. Preserve the supplied Chrome History database and verify its integrity.
2. Identify relevant SQLite tables and fields.
3. Convert Chrome WebKit timestamps to UTC.
4. Examine navigation transitions and previous-visit relationships.
5. Identify relevant URL, communication-related, payment-reference, and download artefacts where present.
6. Construct a chronological timeline based on recorded evidence.
7. Assess the strength and limitations of the evidence without overstating attribution.

3. EVIDENCE ACQUISITION AND INTEGRITY

3.1 Evidence source

The supplied teaching material was identified as the source of the Chrome History database.

mkdir -p evidence working reports screenshots

cp --preserve=timestamps History evidence/History_original

chmod 444 evidence/History_original

cp evidence/History_original working/History

- Original filename: History
- Original SHA-256: b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
- Preserved original location: `evidence/History`
- Working-copy location: `working/History`
- Examination system: Kali Linux
- Examination tools: `sqlite3`



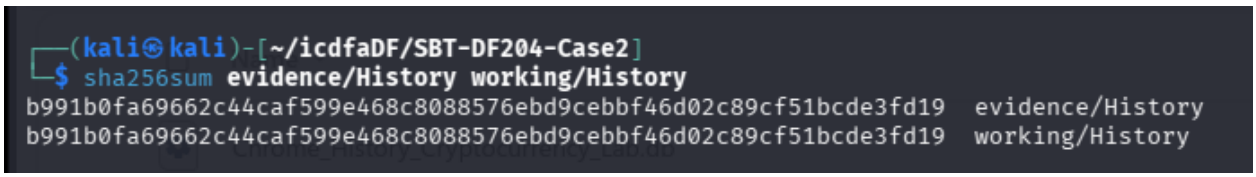
3.2 Preservation procedure

The supplied database was copied into an evidence directory and a separate working copy was created. The preserved original was made read-only, and the examination was conducted using the working copy.

SHA-256 hashes were calculated for the preserved original and working copy. The results were compared to verify that the copy matched the preserved source at the time of verification.

Original SHA-256: b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
Working-copy SHA-256:

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
Hash comparison: Match

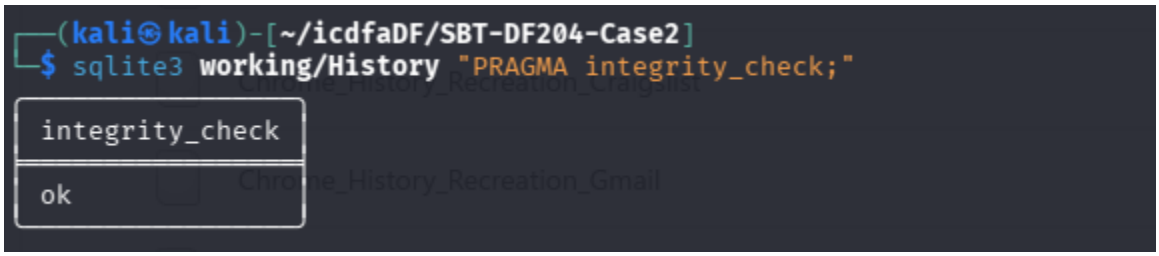


The file metadata, hash output, and source location were documented in the evidence log.

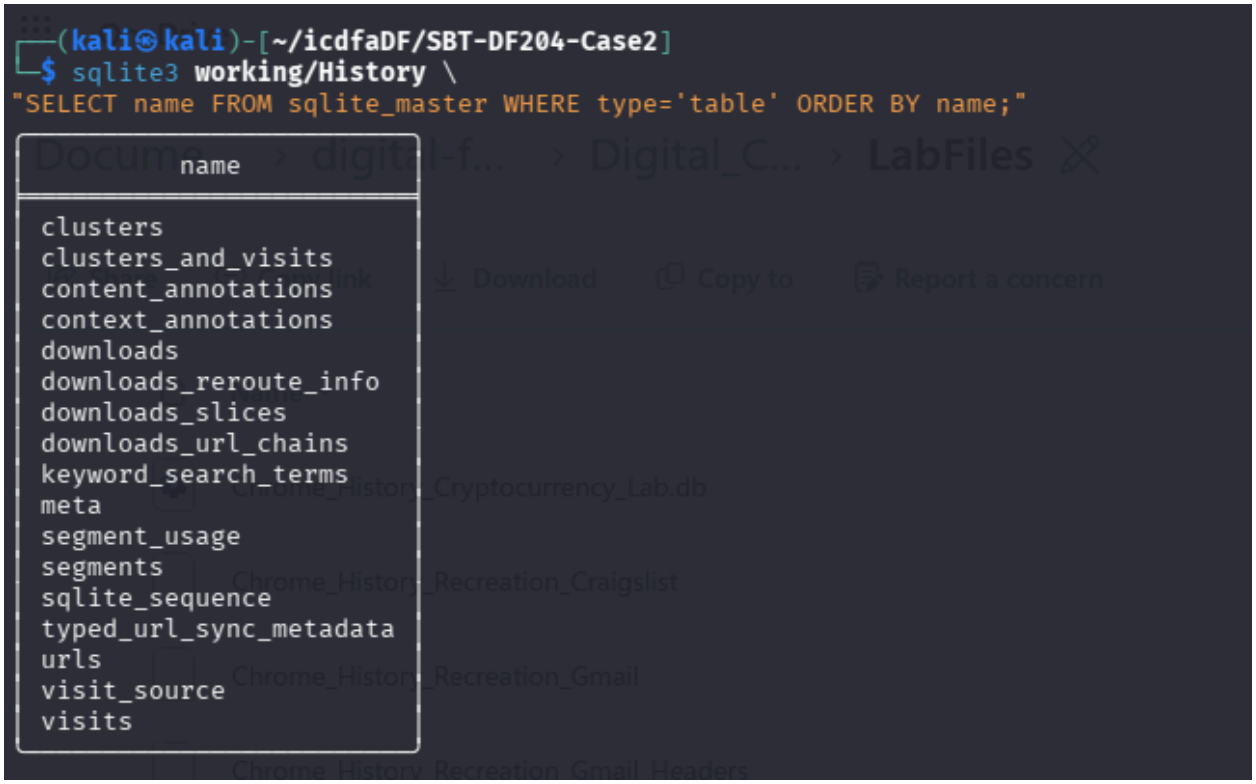
4. DATABASE EXAMINATION METHOD

4.1 Database validation

The file type was checked, and SQLite integrity was tested using `PRAGMA integrity_check`. The result was recorded as ok



The database schema was examined before selecting fields for analysis. The tables reviewed included `urls`, `visits`, `downloads`, and `downloads_url_chains`, where available.



4.2 Timestamp conversion

The database was inspected with SQLite. The relevant tables were urls, visits, downloads, and downloads_url_chains. The urls table stores URL/title metadata and aggregate visit counts. The visits table stores individual visit timestamps and navigation relationships. The downloads table stores download metadata, including target path, start time, and byte counts. The downloads_url_chains table can associate a download with its source/redirect URLs; however, the actual row output from that table was not included in the supplied findings.

Chrome timestamps were converted from WebKit microseconds since 1601-01-01 UTC to UTC using the following query expression:

datetime(v.visit_time / 1000000 - 11644473600, 'unixepoch')

The visit query joined visits.url to urls.id and sorted by visit_time. The navigation review also examined from_visit, the stored transition value, and transition & 255 (the base transition type). The evidence supplied includes base values 5 and 7 in the early search/posting sequence, with many other visits having base value 0. Transition values and parent visit links are interpreted cautiously because redirects and browser navigation qualifiers can affect these fields.

5. FINDINGS

Commands use
 sqlite3 -header -column working/History "
SELECT
    v.id AS visit_id,
    datetime(v.visit_time / 1000000 - 11644473600, 'unixepoch') AS visit_utc,
    u.title,
    u.url,
    v.from_visit,
    v.transition,
    (v.transition & 255) AS transition_base
FROM visits AS v
JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time ASC;
"

5.1 Craigslist listing workflow

Between 13:54:19 and 13:58:31 UTC on 19 April 2022, the history records navigation through Craigslist's general and Baltimore pages, account/login pages, and posting workflow pages. The recorded sequence includes choose type, choose category, create posting, add map/geoverify, edit image, preview, posting confirmation, and manage posting. At 13:58:27 UTC, the browser visited a management URL ending in /manage/7473121658. At 13:58:31 UTC, a public listing URL was visited with the title “cheaper than Rx supplements - health and beauty - by owner - …”.

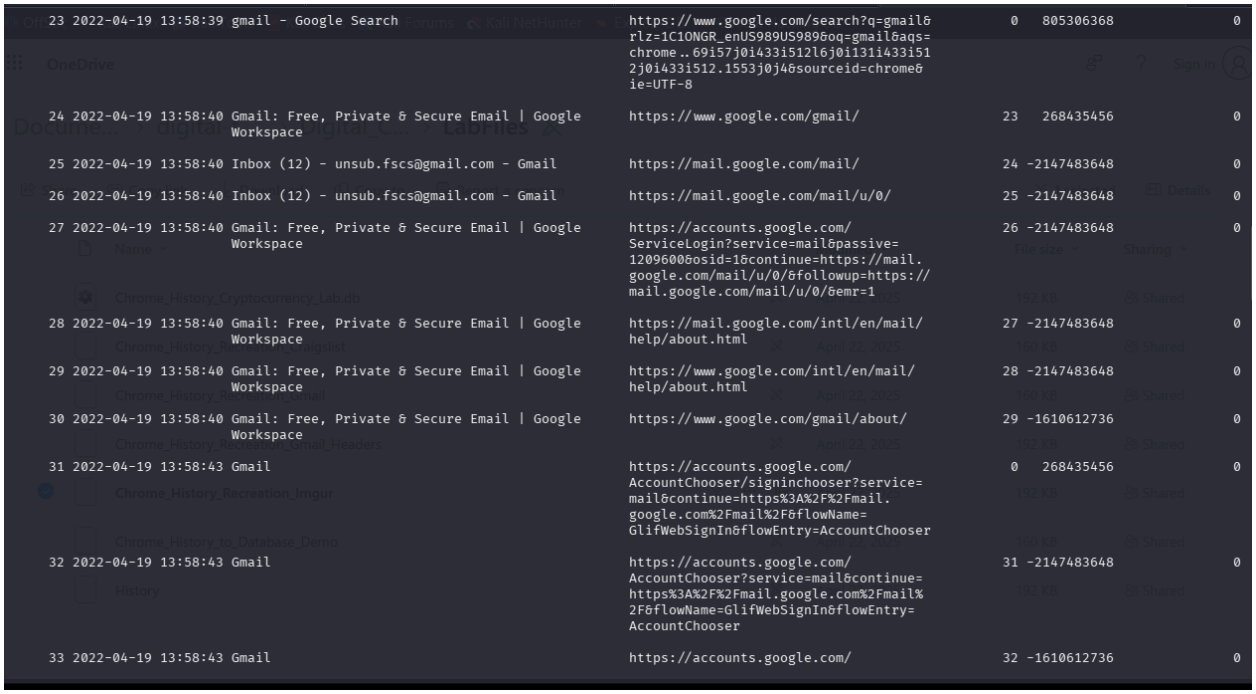
This sequence is consistent with creating or managing a Craigslist post. The History database does not preserve the full listing body in the findings provided, so the title alone cannot establish the product, contents, or legality of the offer.

visit_id	visit_utc	title	url	from_visit	transition	transition_base
1	2022-04-19 13:54:18	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&...	0	805306373	5
2	2022-04-19 13:54:19	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&...	0	805306368	0
3	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://www.craigslist.org/	2	268435456	0
4	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://geo.craigslist.org/	3	-2147483648	0
5	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://baltimore.craigslist.org/	4	-1610612736	0
6	2022-04-19 13:54:22	craigslist account	https://accounts.craigslist.org/login/home	5	268435456	0
7	2022-04-19 13:54:22	craigslist - account log in	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L	6	-1610612736	0
8	2022-04-19 13:54:39	craigslist account	https://accounts.craigslist.org/login/home	7	805306375	7
9	2022-04-19 13:55:03	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://www.craigslist.org/	8	268435456	0
10	2022-04-19 13:55:03	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://baltimore.craigslist.org/	9	-1610612736	0

5.2 Gmail and message-related records

The history records Gmail access associated with the account label unsub.fscs@gmail.com. At 14:56:47 UTC, a Gmail history entry has the title “cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com”. Additional Gmail URLs at approximately 14:04 and 15:02 contain inbox and compose fragments associated with a message title “cheaper than Rx supplements - order”.

These are browser title/URL artifacts, not the email contents. They do not identify the sender, recipient, full message text, or whether any communication was successfully sent.



23	2022-04-19 13:58:39	gmail - Google Search	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmail&aqs=chrome..69i57j0i433i512l6j0i131i433i512j0i433i512.1553j0j4&sourceid=chrome&ie=UTF-8	0	805306368	0
24	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email | Google Workspace	https://www.google.com/gmail/	23	268435456	0
25	2022-04-19 13:58:40	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/	24	-2147483648	0
26	2022-04-19 13:58:40	Inbox (12) - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/	25	-2147483648	0
27	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email | Google Workspace	https://accounts.google.com/ServiceLogin?service=mail&passive=1209600&osid=1&continue=https://mail.google.com/mail/u/0/&followup=https://mail.google.com/mail/u/0/&emr=1	26	-2147483648	0
28	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email | Google Workspace	https://mail.google.com/intl/en/mail/help/about.html	27	-2147483648	0
29	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email | Google Workspace	https://www.google.com/intl/en/mail/help/about.html	28	-2147483648	0
30	2022-04-19 13:58:40	Gmail: Free, Private & Secure Email | Google Workspace	https://www.google.com/gmail/about/	29	-1610612736	0
31	2022-04-19 13:58:43	Gmail	https://accounts.google.com/AccountChooser/signinchooser?service=mail&continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&flowName=GlifWebSignIn&flowEntry=AccountChooser	0	268435456	0
32	2022-04-19 13:58:43	Gmail	https://accounts.google.com/AccountChooser?service=mail&continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&flowName=GlifWebSignIn&flowEntry=AccountChooser	31	-2147483648	0
33	2022-04-19 13:58:43	Gmail	https://accounts.google.com/	32	-1610612736	0

5.3 Download record

Command use

sqlite3 -header -column working/History "

SELECT

id AS download_id,

target_path,

datetime(start_time / 1000000 - 11644473600, 'unixepoch') AS start_utc,

received_bytes,

total_bytes,

state,

danger_type,

mime_type

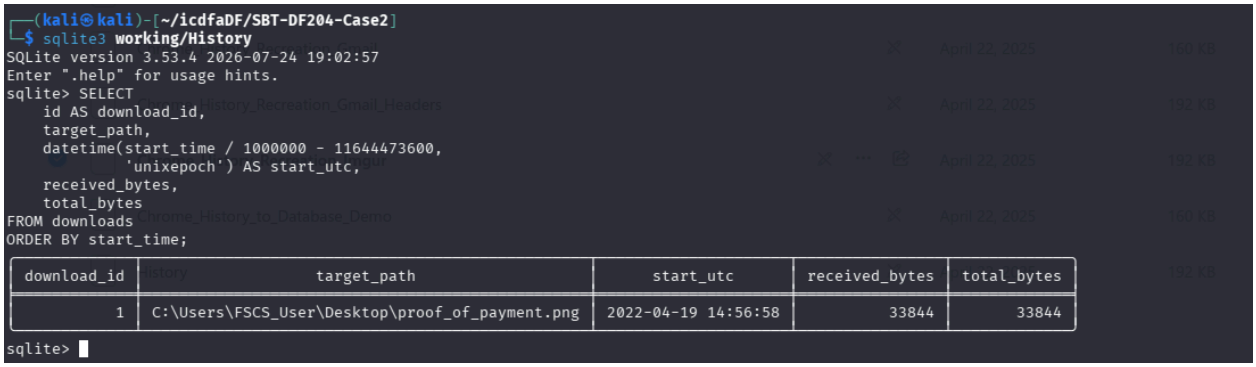
FROM downloads

ORDER BY start_time;

"

Field	Value
Download ID	1
Target path	**c:**\users\FSCS_User\Desktop\proof_of_payment.png
Start time (UTC)	2022-04-19 14:56:58
Received bytes	33,844
Total bytes	33,844

The Downloads table contains one recorded download. The received byte count equals the total byte count, which is consistent with the download reaching its recorded size. The image itself was not supplied for inspection in the findings available to this report. The filename “proof_of_payment.png” is not proof that a payment occurred. The actual download URL-chain rows were not provided, so the source URL cannot be confirmed from the supplied output.



5.4 Blockchain and cryptocurrency-related browsing

Command use

sqlite3 -header -column working/History "

SELECT

v.id AS visit_id,

datetime(v.visit_time / 1000000 - 11644473600, 'unixepoch') AS visit_utc,

u.title,

u.url,

```
    v.from_visit,

    v.transition

FROM visits AS v

JOIN urls AS u ON v.url = u.id

WHERE u.url LIKE
'%517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5%'

   OR u.title LIKE
'%517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5%'

ORDER BY v.visit_time;

"
```

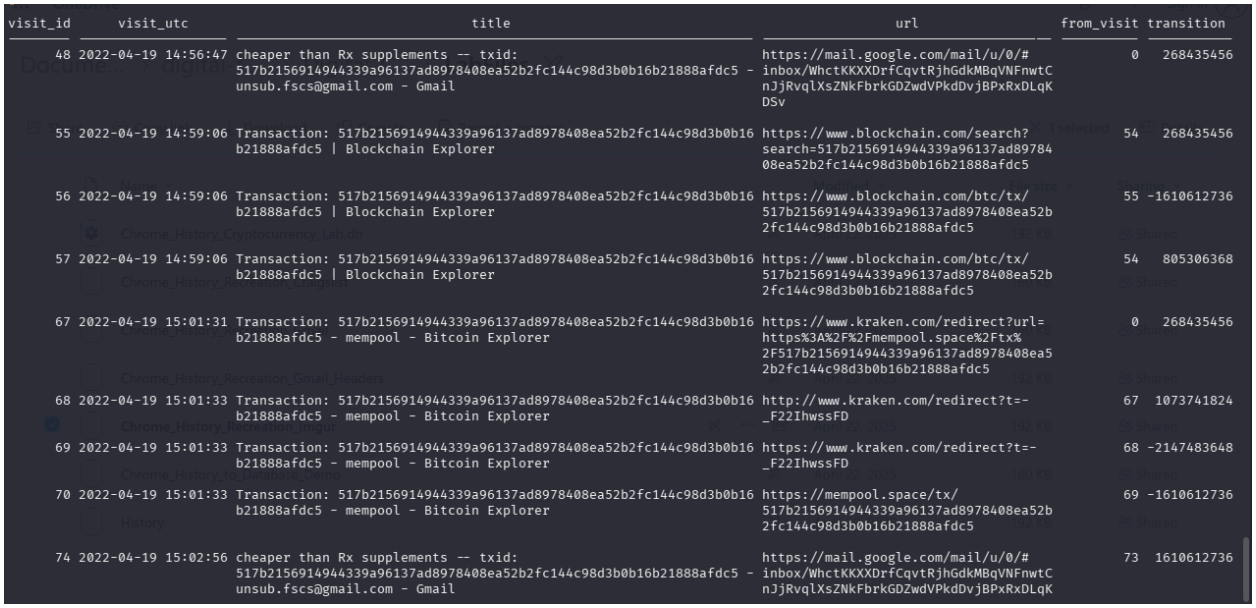
Between 14:59:01 and 15:01:33 UTC, the browser history records visits to Blockchain.com, a transaction page, a Bitcoin address page, Kraken pages, and a Mempool transaction page. The transaction identifier appearing in the relevant URLs and Gmail title is:

517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5

The Bitcoin address recorded in the history is:

38RcsURWCDmCbYocmsZCnGz1FtD7x477mt

The history includes a Kraken funding page titled “41,543.80 USD - Kraken - Funding - Overview” at 15:01:21 UTC and a ledger page titled “41,563.70 USD - Kraken - History - Ledger” at 15:02:05 UTC. These values are page titles recorded by Chrome; without the underlying account records, their meaning, accuracy, ownership, and relation to the transaction cannot be independently verified. Visiting a transaction/address page does not by itself prove that the browser user controlled the address or sent/received funds.



visit_id	visit_utc	title	url	from_visit	transition
48	2022-04-19 14:56:47	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqKDSv	0	268435456
55	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer	https://www.blockchain.com/search?search=517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	54	268435456
56	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer	https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	55	-1610612736
57	2022-04-19 14:59:06	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer	https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	54	805306368
67	2022-04-19 15:01:31	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%2F517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	0	268435456
68	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	http://www.kraken.com/redirect?t=-_F22IhwssFD	67	1073741824
69	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	https://www.kraken.com/redirect?t=-_F22IhwssFD	68	-2147483648
70	2022-04-19 15:01:33	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5	69	-1610612736
74	2022-04-19 15:02:56	cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com - Gmail	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqK	73	1610612736

Download URL-chain records

Command use

sqlite3 -header -column working/History "

SELECT

    id AS download_id,

    chain_index,

    url

FROM downloads_url_chains

ORDER BY id, chain_index;

"



6. RECONSTRUCTED TIMELINE

Command use

sqlite3 -header -column working/History "

SELECT

v.id AS visit_id,

datetime(

v.visit_time / 1000000 - 11644473600,

'unixepoch'

) AS timestamp_utc,

u.title AS page_title,

u.url AS page_url,

v.from_visit AS referring_visit_id,

v.transition AS raw_transition,

(v.transition & 255) AS transition_base,

ROUND(v.visit_duration / 1000.0, 2) AS duration_ms

FROM visits AS v

JOIN urls AS u

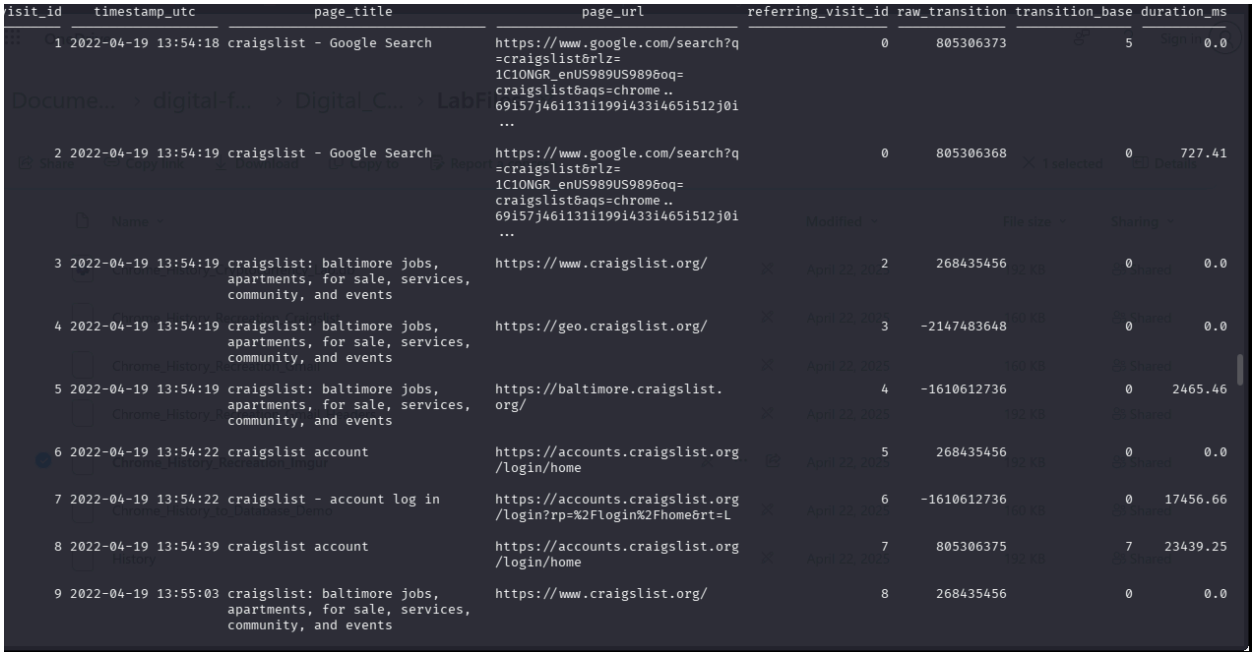
ON v.url = u.id

ORDER BY v.visit_time ASC;

"

UTC time (19 April 2022)	Event summary
13:54:18–13:55:03	Google search and Craigslist general/Baltimore pages; account/login pages
13:55:05–13:58:22	Craigslist posting workflow: choose type/category, edit, map verification, image step, preview, confirmation
13:58:27	Craigslist manage-posting URL ending /manage/7473121658
13:58:31	Public listing URL titled “cheaper than Rx supplements”
13:58:38–14:05:59	Google/Gmail pages, sign-in flow, inbox, and message/compose-related URLs
14:56:47	Gmail history title references “cheaper than Rx supplements” and the transaction ID
14:56:52	Imgur page dTgrkP7 visited
14:56:58	Downloads table records proof_of_payment.png; 33,844 of 33,844 bytes
14:58:59–14:59:21	Search for blockchain explorer; transaction page and Bitcoin address page visited
14:59:38–15:01:21	Search for Kraken; sign-in/device approval, trade/instant, and funding pages
15:01:31–15:01:33	Kraken redirect chain leads to Mempool transaction page for the same transaction ID
15:02:05	Kraken ledger page visited
15:02:32–15:02:56	Gmail compose-related URLs revisited; final recorded title references the transaction ID

The times above are UTC converted from Chrome WebKit timestamps. The sequence is a reconstruction of recorded browser events, not proof that each page fully loaded or that a corresponding real-world action was completed.



visit_id	timestamp_utc	page_title	page_url	referring_visit_id	raw_transition	transition_base	duration_ms
1	2022-04-19 13:54:18	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i...	0	805306373	5	0.0
2	2022-04-19 13:54:19	craigslist - Google Search	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i...	0	805306368	0	727.41
3	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://www.craigslist.org/	2	268435456	0	0.0
4	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://geo.craigslist.org/	3	-2147483648	0	0.0
5	2022-04-19 13:54:19	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://baltimore.craigslist.org/	4	-1610612736	0	2465.46
6	2022-04-19 13:54:22	craigslist account	https://accounts.craigslist.org/login/home	5	268435456	0	0.0
7	2022-04-19 13:54:22	craigslist - account log in	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L	6	-1610612736	0	17456.66
8	2022-04-19 13:54:39	craigslist account	https://accounts.craigslist.org/login/home	7	805306375	7	23439.25
9	2022-04-19 13:55:03	craigslist: baltimore jobs, apartments, for sale, services, community, and events	https://www.craigslist.org/	8	268435456	0	0.0

7. Navigation and Transition Interpretation

The visits query linked each visit to its URL record and displayed from_visit. In the Craigslist posting sequence, the parent links form a largely continuous chain from the regional Craigslist pages through the posting workflow. The transition base value is obtained with (transition & 255). Values of 5 occur on Google search visits and values of 7 occur on several posting workflow visits in the supplied output. Many other entries have base value 0. These values are reported as recorded; they should not be treated as conclusive proof of a particular human action. Redirect qualifiers, browser internals, and incomplete chains may explain some visit relationships.

8. Assessment of the Evidence and Attribution

Proposition	**Support in the supplied browser evidence**	**Assessment**
A Craigslist listing workflow occurred	Multiple sequential posting, preview, confirmation, and management URLs; listing title	Strong support for browsing through the workflow; actual post content needs corroboration
The listing related to “Rx supplements”	Public URL/title and Gmail titles use that phrase	Supported as a recorded title; contents and legality are not established
A file called	Downloads row with target path and	Strong support that Chrome recorded

proof_of_payment.png was downloaded	matching received/total byte counts	the download; image contents and payment are unverified
A transaction ID was researched or referenced	Gmail title plus Blockchain.com and Mempool URLs include the same ID	Strong support for related browsing/reference activity; transaction outcome not established
Funds were transferred through Kraken or the address	Kraken page titles and transaction/address browsing	Not established from browser history alone
A specific individual performed the activity	Profile/account labels and a Windows-style target path	Not established; profile and device/user identity require independent attribution

The evidence supports a connected pattern of browser activity involving a Craigslist listing, Gmail message titles, a download labelled as proof of payment, and cryptocurrency-related pages. The pattern is relevant to the suspected case, but it remains circumstantial. No conclusion that drug trafficking or a completed payment occurred should be made without corroborating evidence

9. Limitations and Further Examination

• The source device, forensic acquisition procedure, original custodian, and chain-of-custody documentation were not supplied.

• The downloaded image proof_of_payment.png was not available for content examination or independent hashing. • The actual downloads_url_chains records were not included, so the download source URL is unknown.

• The full Craigslist listing body, associated image(s), and seller/account records were not provided.

• The full Gmail message bodies, headers, sender/recipient details, and sent-mail evidence were not provided.

• No independently verified blockchain transaction details, wallet ownership evidence, or Kraken account/export records were supplied.

• Chrome history can show URLs and titles recorded by a browser, but does not by itself identify the human operator or prove that a transaction completed.

• The database may not contain all browsing activity; absence of a record does not prove an event did not occur.

10. CONCLUSION

The Chrome History database records a sequence on 19 April 2022 that begins with Craigslist navigation and posting pages, including a listing titled “cheaper than Rx supplements.” The same browser history contains Gmail titles referencing that phrase and a transaction identifier, a download record for proof_of_payment.png, and subsequent visits to Blockchain.com, a Bitcoin address, Kraken, and Mempool pages associated with the same transaction identifier. The database and working copy have matching SHA-256 hashes, and the SQLite integrity check returned “ok.” Overall assessment: the browser artifacts provide meaningful investigative leads and support that these pages and download were recorded in the Chrome profile. They do not independently prove a drug sale, payment, ownership of the cryptocurrency address, or the identity of the person operating the browser. Those propositions require corroboration from the actual image, email contents and headers, listing content, exchange records, blockchain analysis, and device/user attribution.